

Development and Performance Validation of a Profile HMM for the Annotation of Kunitz-type Domains

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Abstract

Given the growing importance of Kunitz-type serine protease inhibitors across diverse organisms and their potential in protein engineering for therapeutic applications, accurate domain annotation has become essential. This project aims to build a high-quality Hidden Markov Model (HMM) by curating a non-redundant training set of structural monodomains. This approach ensures the minimization of classification errors and provides a robust strategy for the reliable identification of Kunitz domains in protein sequences.

Supplementary material: <https://github.com/Francesca-Crispino/HMM-Kunitz-domain>

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1. INTRODUCTION

1.1 Kunitz-type protease inhibitor domain: biological introduction

The Kunitz-type inhibitor family represents one of the best-characterized groups of serine protease inhibitors and it is regarded a key model for studying protein-protein interactions and protein structural stability. First identified in the 1930s through the isolation of the bovine pancreatic trypsin inhibitor (BPTI), the Kunitz domain is now recognized as a widely structural motif distributed across organisms ranging from simple invertebrates to mammals [1]. Kunitz-type inhibitors are classified in the MEROPS database [2] into two main families, I2 and I3. Family I2 includes mainly animal-derived inhibitors that are specific for S1 proteases, particularly serine proteases. Family I3 consists mainly of plant-derived inhibitors, which primarily target S1 serine proteases but may also show inhibitory activity against cysteine (C1) and, less commonly, aspartic (A1) proteases. Gene sharing and gene duplication have been proposed as the principal evolutionary mechanisms underlying the remarkable functional diversity of Kunitz-domain inhibitors. Kunitz-type domains are found in a broad variety of proteins and biological contexts, including toxins from venomous animals such as snakes, spiders, and scorpions; mammalian inter- α -trypsin inhibitors; the amyloid β -protein associated with Alzheimer's disease in humans and tissue factor pathway inhibitor (TFPI) [3]. Their extensive distribution and structural variability strongly suggest an evolutionary origin driven by exon shuffling, involving the recombination and incorporation of genetic elements derived from distinct ancestral proteins [4] [5]. Structurally, the Kunitz domain is

a small cysteine-rich polypeptide (3-20kDa) typically composed of 50-70 amino acid residues, which may contain a single mono-inhibitory domain or multiple in tandem domains, within a single polypeptide chain. Its exceptional stability due to from a conserved α/β fold stabilized by three highly conserved disulfide bridges arranged in the characteristic connectivity pattern C1–C6, C2–C4, and C3–C5. This covalent framework confers resistance to thermal denaturation and proteolytic degradation. In many Kunitz inhibitors, the abundance of basic amino acids, particularly arginine and lysine residues, also contributes to a relatively basic isoelectric point [6]. The inhibitory mechanism of Kunitz domains is based on a principle that may be described as “*flexibility within structural rigidity*” The inhibitory mechanism of Kunitz domains relies on a flexible reactive loop within a structurally stable scaffold. This exposed loop (about 8–19 residues) fits into the active site of target proteases. In BPTI, for example, the P1 residue Lys15 interacts with Asp189 in trypsin's S1 pocket. Inhibitor specificity is mainly determined by the P1 residue, typically lysine or arginine, which inserts into the S1 specificity pocket. By mimicking a natural substrate, the reactive loop forms a highly stable, pseudo-irreversible protease-inhibitor complex. However, unlike true substrates, it is not cleaved because the disulfide-stabilized scaffold restricts loop flexibility and prevents full hydrolysis, effectively blocking protease activity [7]. Functionally, Kunitz-domain inhibitors participate in numerous physiological and pathological processes, including host defense against microbial infection, regulation of blood coagulation and fibrinolysis, modulation of inflammatory

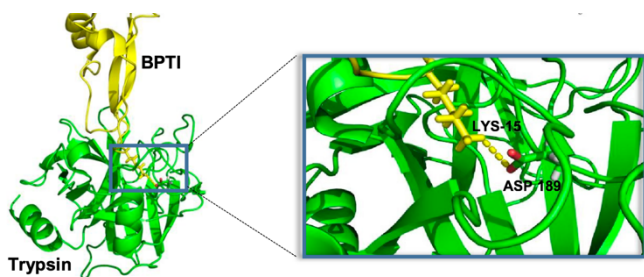


Figure.1 Trypsin in complex with BPTI (PDB: 4Y0Y) showing P1 residue (K15 in BPTI) interacting with the Asp189 at the bottom of the S1 pocket of Trypsin.

1.2 Hidden Markov Model

Hidden Markov Model (HMMs) is a probabilistic model designed to describe systems in which observable events depend on underlying hidden factors that cannot be directly observed: the observable events are referred to as *symbols* whereas the hidden underlying factors are called *states*. Because the observed symbols are probabilistically generated from hidden states, HMMs are often described as doubly embedded stochastic processes: a hidden process describing transitions between states and an observable process generating symbols according to state-specific probability distributions. The hidden states follow a Markov chain, meaning that the probability of transitioning to a new state depends only on the current state. In protein sequence analysis, Hidden Markov Model profile is particularly useful for detecting conserved patterns characteristic of specific protein families. These patterns reflect the conservation of residues that are often functionally or structurally important. The construction of a profile HMM begins with the collection of homologous protein sequences, which are aligned to generate a multiple sequence alignment and a corresponding sequence profile, which describes the conservation level of residues at each position within the protein family. A profile HMM represents each position of the alignment using a series of states associated with specific probabilities. The most important of these are *match states*, which correspond to conserved positions in the alignment. Each match state is associated with a set of *emission probabilities*, representing the likelihood of observing each amino acid residue at that specific position. Conserved residues typically display high emission probabilities, whereas less common residues are assigned lower probabilities. In addition to match states, realistic profile HMMs also

responses, and control of extracellular proteolytic activity through the inhibition of serine proteases.

include *insertion* and *deletion* states, which account for gaps and variable regions commonly observed in biological sequences. These probabilities allow profile HMMs to capture both sequence conservation and structural variability, making them highly effective tools for protein family classification, homology detection, and functional annotation. [8]

2. Materials and Methods

2.1 Training Set Selection

2.1.1 Database querying and initial filtering

The initial dataset was retrieved from the RCSB Protein Data Bank (PDB) [9] using an advanced search strategy specifically designed to isolate protein structures containing the Kunitz domain, as defined by the protease inhibitor family Pfam identifier PF00014. To ensure structural reliability, only entries with a refinement resolution of ≤ 3.5 Å were included. Following the search, a customized tabular report was generated to extract relevant metadata for each entry, including the PDB entry ID, Chain identifier and amino acid sequence. The tabular report is available in supplementary materials. The resulting dataset subsequently underwent an initial computational filtering step in which polymer entity sequences were restricted to a length range of 40-80 amino acid residues, corresponding to the canonical size of the Kunitz monodomain while excluding short fragments and larger multidomain proteins.

2.1.2 Sequence redundancy Reduction

The dataset was subsequently parsed using a Bash script to generate a refined collection of unique sequence identifiers and corresponding amino acid sequences. To minimize statistical bias during Hidden Markov Model (HMM) construction and avoid overrepresentation of closely related proteins, a sequence clustering step was performed. The MMseqs2 [10] web server was employed to group highly similar sequences into clusters using a greedy set cover algorithm. Clustering parameters were stringently selected to ensure a high degree of sequence and structural homogeneity within each cluster, using: a sequence identity threshold of 95% and a minimum alignment coverage of 90%. From each resulting cluster, a single representative sequence was retained. This curated nonredundant dataset preserves the structural and evolutionary diversity of the Kunitz domain fold while reducing the

redundancy intrinsic to the PDB dataset, thereby providing a balanced and reliable foundation for subsequent HMM construction.

2.2 Multiple Structural Alignment

The list of the representative chain list was generated using the *get_chains.py* script (supplementary materials), which was specifically developed to extract monodomain Kunitz coordinates from the original PDB structures. The processed chain structures were subsequently submitted to mTM-align [11]; a tool designed for multiple protein structural alignment. Unlike conventional sequence-based alignment approaches, mTM-align relies on structural similarity metrics, particularly the Template Modeling score (TM-score), to identify the optimal topological correspondence between protein structures. The resulting structural alignment was then converted into a FASTA-formatted file, referred to as *kunitz.ali*, which served as the seed alignment for profile construction and subsequent Hidden Markov Model generation.

2.2.3 Profile-HMM construction and visualization

The seed MSA was used as input for the *hmmbuild* module of the HMMER 3.4 [12] suite. This procedure generated a profile Hidden Markov Model (profile-HMM). To evaluate the information content and positional conservation of residues, the resulting profile-HMM was visualized using the Skyline web server [13]. The generated HMM logo provided a graphical representation of residue conservation, amino acid frequency, and biochemical property enrichment at each position of the Kunitz domain.

2.3 Selection of the Test Set

To evaluate the predictive performance of the profile-HMM, a benchmark dataset containing Kunitz-domain proteins (positive) and non-Kunitz proteins (negative) sequences was assembled from the reviewed UniProtKB/Swiss-Prot database [14]. The positive set included all proteins annotated with the Pfam Kunitz domain (PF00014; n = 398), while the remaining reviewed entries (574,229 proteins) were used as the negative control dataset. This comprehensive negative background enabled reliable estimation of false discovery rates and optimization of E-value thresholds.

2.3.2 Data Cleaning with Blast

To ensure an unbiased validation procedure, sequences highly like those used during model

training were excluded from the benchmark dataset. Omologs sequences were identified using the BLASTP [15] algorithm. The 19 structural representative sequences used for HMM construction was compiled into a local reference database against which the positive dataset was queried. A stringent similarity filter was then applied: any sequence sharing $\geq 95\%$ sequence identity with a training seed, was removed from the test set.

2.4 Model testing

To evaluate the predictive performance of the profile-HMM and identify the optimal classification threshold, the model was tested against the curated benchmark datasets using a two-fold cross-validation strategy. The *hmmsearch* command from was used to scan both the positive and negative datasets. The relative output files were parsed to extract the sequence identifiers and the E-value and each sequence was assigned a binary label: 1 for Kunitz-domain proteins and 0 for non-Kunitz proteins. Was employed a specific approach for handling non-matches, to preserve the statistical integrity of the benchmark set: sequences from the negative dataset producing no alignment were reintroduced into the dataset and assigned a penalty E-value of 100, ensuring their correct classification as non-members during threshold optimization and downstream performance analyses.

2.4.3 E-value optimization and MCC

To determine the optimal E-value, threshold for sequence classification, an optimization procedure was performed on two subsets (*kunitz_set1* and *kunitz_set2*) generated through random splitting and sorting of the complete benchmark dataset. Performance evaluation was carried out using the *performance.py* script (supplementary materials). The primary metric used to rank threshold performance was the Matthews Correlation Coefficient (MCC), defined as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

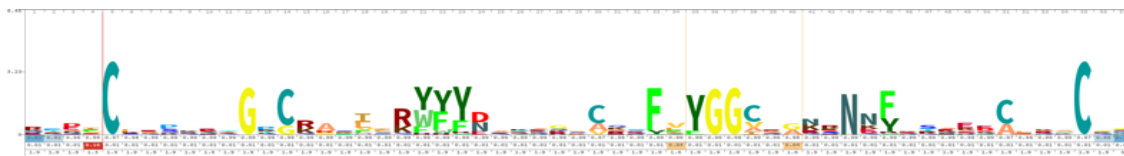


Figure 2.HMM profile logo downloaded from SkyAlign

The MCC metric provides a balanced evaluation of binary classification performance even in the presence of substantial class imbalance, such as the limited number of positive Kunitz sequences compared with the extensive negative Swiss-Prot background. By identifying the E-value threshold that maximized the MCC across both cross-validation subsets, the optimization procedure enabled the selection of a cutoff that achieved the best balance between sensitivity (minimization of false) and specificity (minimization of false positives).

3. RESULTS

3.1 Training set selection, MSA.

The advanced search of the RCSB Protein Data Bank retrieved 135 Kunitz-domain polymer entries, which were reduced to 102 sequences after filtering. Redundancy reduction with MMseqs2 identified 19 sequence clusters, from which representative structures were selected. Structural alignment revealed high conservation of the canonical Kunitz fold, and two furviant structures (5YW1A and 1D0DA) were excluded after an initial alignment. The final alignment showed an RMSD of 0.79 Å and a TM-score of 0.90, confirming strong structural similarity while preserving sufficient sequence diversity for robust profile-HMM construction.

3.3 Evaluation of Hidden Markov Model

The Hidden Markov Model resulting model showed a match-state length of 57 residues, consistent with the canonical size of the Kunitz monodomain. The resulting sequence HMM logo performed with SkyAlign highlights the six conserved cysteine residues responsible for the characteristic disulfide-bond architecture of the Kunitz fold (Figure2).

3.4 Classification and Performance results

Threshold optimization showed slight differences between the two subsets. Subset 1 achieved an MCC of 0.99 at an E-value threshold of 10^{-7} , whereas Subset 2 reached an MCC of 0.99 at 10^{-3} . A threshold of 10^{-5} was therefore selected to balance sensitivity and specificity across

the full dataset, justified by the MCCvsE-value curves (Figure3); in fact, at this threshold, the model showed strong discriminative performance: Set 1 yielded 191 true positives, 7 false positives, and no false negatives, while Set 2 identified 187 true positives with no false positives and 4 false negatives (Figure 4-5).

The ROC curve analyses further supported these results showing consistently strong performance across both validation subsets, with high sensitivity maintained at very low false-positive rates (Figure6).

4. Conclusions

The primary objective of this study was to develop a Hidden Markov Model capable of providing reliable and sensitive annotation of the Kunitz domain, starting from a structurally defined set of representative proteins. Stringent structural filtering criteria and manual curation of the dataset enabled the construction of a model that effectively captures the core biological characteristics of the Kunitz protein family. However, some like Kunitz-type proteins, BLI-5 from *Caenorhabditis elegans* (O62247) and *Haemonchus contortus* (D3GGZ8), were classified as false negatives, likely representing highly divergent homologs near the limits of the model's statistical confidence. Despite these edge cases, the model combined high specificity with robust sensitivity across a large and diverse dataset.

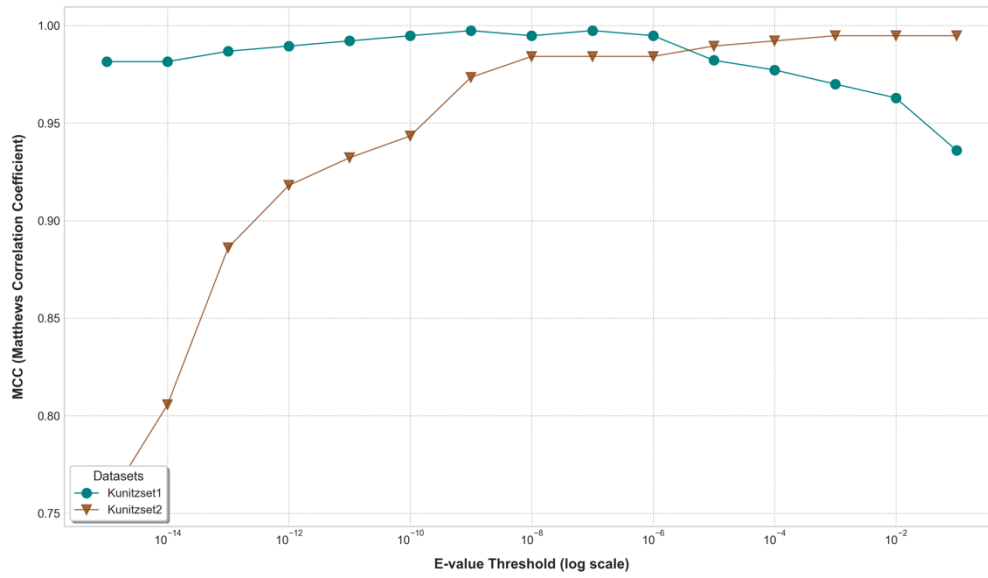


Figure 3. MCCvsEvalue Threshold.

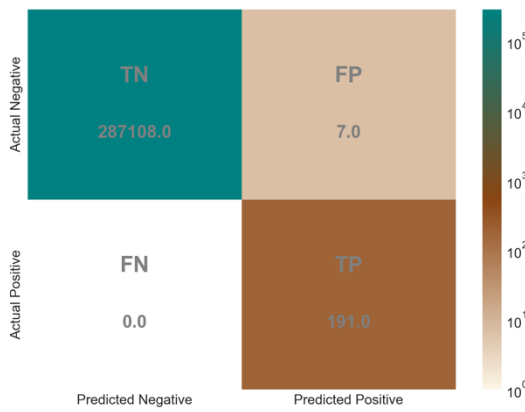


Figure 4. Confusion Matrix Kunitz_set1 (Th: 10^{-5})

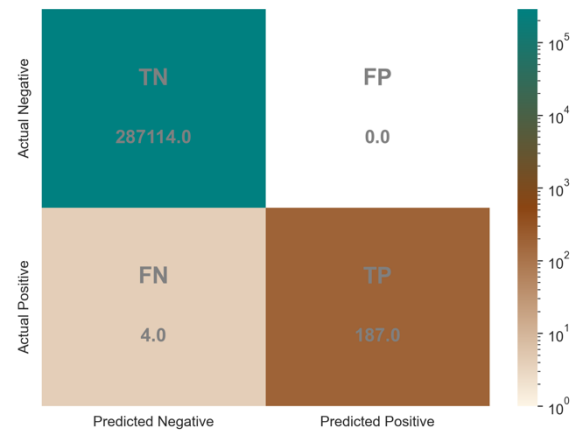


Figure 5. Confusion Matrix Kunitz_set2 (Th: 10^{-5})

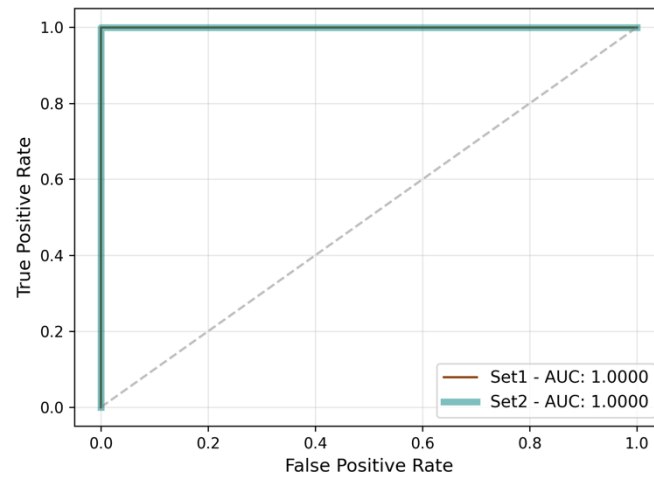


Figure 6. ROC curve for HMM Cross-validation

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